astronomers tended to espouse this view, when they thought at all about the issue of the final fates of stars. (By and large, however, they didn't think about the issue. There were no observational data pushing them to think about it; and the observational data that they were gathering on other kinds of objects—normal stars, nebulas, galaxies—were so rich, challenging, and rewarding as to absorb the astronomers' full attention.)

Today, in the 1990s, we know that heavy stars *do* eject enormous amounts of mass as they age and die; they eject so much, in fact, that most stars born with masses as large as 8 Suns lose enough to wind up in the white-dwarf graveyard, and most born between about 8 and 20 Suns lose enough to wind up in the neutron-star graveyard. Thus, nature seems *almost* to protect herself against black holes. But not quite: The preponderance of the observational data suggest (but do not yet firmly prove) that most stars born heavier than about 20 Suns remain so heavy when they die that their pressure provides no protection against gravity. When they exhaust their nuclear fuel and begin to cool, gravity overwhelms their pressure and they implode to form black holes. We shall meet some of the observational data suggesting this in Chapter 8.

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There is much to be learned about the nature of science and scientists from the neutron-star and neutron-core studies of the 1930s.

The objects that Oppenheimer and Volkoff studied were Zwicky's neutron stars and not Landau's neutron cores, since they had no surrounding envelope of stellar matter. Nevertheless, Oppenheimer had so little respect for Zwicky that he declined to use Zwicky's name for them, and insisted on using Landau's instead. Thus, his article with Volkoff describing their results, which was published in the 15 February 1939 issue of the *Physical Review,* carries the title "On Massive Neutron Cores." And to make sure that nobody would mistake the origin of his ideas about these stars, Oppenheimer sprinkled the article with references to Landau. Not once did he cite Zwicky's plethora of prior neutron-star publications.

Zwicky, for his part, watched with growing consternation in 1938 as Tolman, Oppenheimer, and Volkoff pursued their studies of neutronstar